Best practices Guide

Strengthening Governance and Optimizing Enterprise Architecture with SAP LeanIX

Level 1 2 3

Recommendation #1

Use cases

Context

With your recent Enterprise Architecture initiatives with SAP LeanIX, you now have a clear view of your As-Is architecture across the different layers of architecture. You can now lead your **transformation projects** by defining your **To-Be Architecture**. On the other hand, we can go further in managing risk management by deepening our understanding of our **in-house developed applications**. Finally, in a context where Artificial Intelligence is an integral part of your architecture, SAP LeanIX will allow you to gain **transparency on the use of Artificial Intelligence** within your company.

Best Practices

- **Get closer to new stakeholders**. Enterprise architecture is the bridge between strategy and IT and in order to do so, it should address all the main stakeholders. That is Software developers, Solutions Architects, business process managers, Infosec & Compliance officers, CIO and CTO, and finally IT program managers. See here how architecture can answer their key questions.
- Focus on **defining**, **prioritizing and validating the next use cases**. Make sure these use cases meet your company's primary goals and by putting them in place, it will naturally enable you to develop the use and adoption of SAP LeanIX within your organization.

	"Traditional" EA	"Collaborative" EA	"Application Intelligence" Predictive EA
People involved	• CIOs • Top Management	• + CTO • + IT Product Mgr • + IT Project Mgr	• + SW Developer • + IT Operations
Use Cases	• Documentation • Rationalization • Standards & Governance	• Digital Transformation • M&A Harmonization • Tech. Risk Mgmt	• Cloud Transformation • Microservice Adoption • IoT Enablement
User Needs	• Generic Repository • Meta-Modelling • Scenario Analysis	• Easy UI / UX • Visualization • Collaboration • Extendable Repository	• Integrations • Automation • Machine Learning • Recommendations

Recommendation #1

Use cases – Transformations

WHAT?

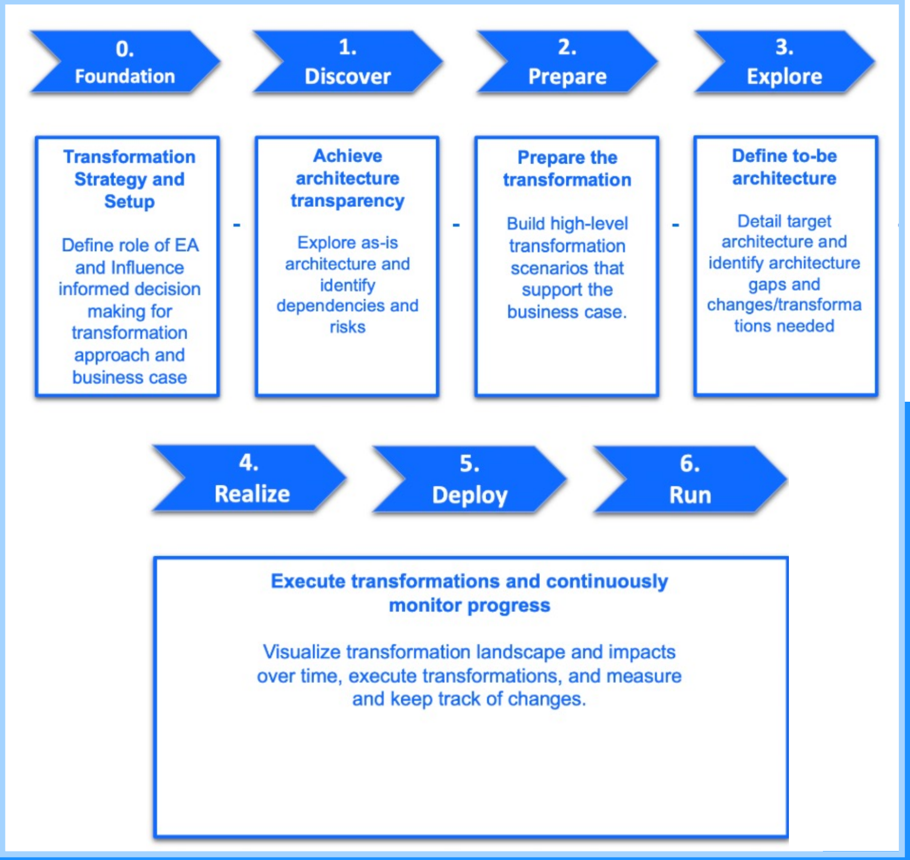
Digital transformations aim to modernize, consolidate, or upgrade core technology systems. This includes transitioning to cloud-based solutions, merging systems, or optimizing costs.
The aim is to support strategic objectives and provide the necessary flexibility to **adapt to changing business needs**.

WHY?

Digital transformations are extremely complex, requiring important architectural decisions. Enterprise architecture management is crucial to managing this complexity. The main challenges facing a transformation project are the lack of shared understanding of **affected processes and technologies**, insufficient awareness of transformation choices and their impacts, limited access to relevant data, and difficulties **tracking post-project impact** for continuous optimization. As an Enterprise Architect, you play a pivotal role in solving these challenges and ensuring a successful transformation with SAP LeanIX.

HOW?

SAP LeanIX supports the transformation by facilitating the **definition of the target architecture** encompassed in those three main steps: **plan, decide, and execute**. *Planning* involves **setting goals** and turning initiatives into actions. *Decide* is about **segmenting initiatives** into executable projects and creating a roadmap. *Executing* requires **continuous monitoring of progress** and regular communication to ensure a successful and secure implementation.
Find our step-by-step guide to transforming your ERP with support from SAP LeanIX.



Recommendation #1
Use Cases – Self-Built Software Discovery

WHAT?

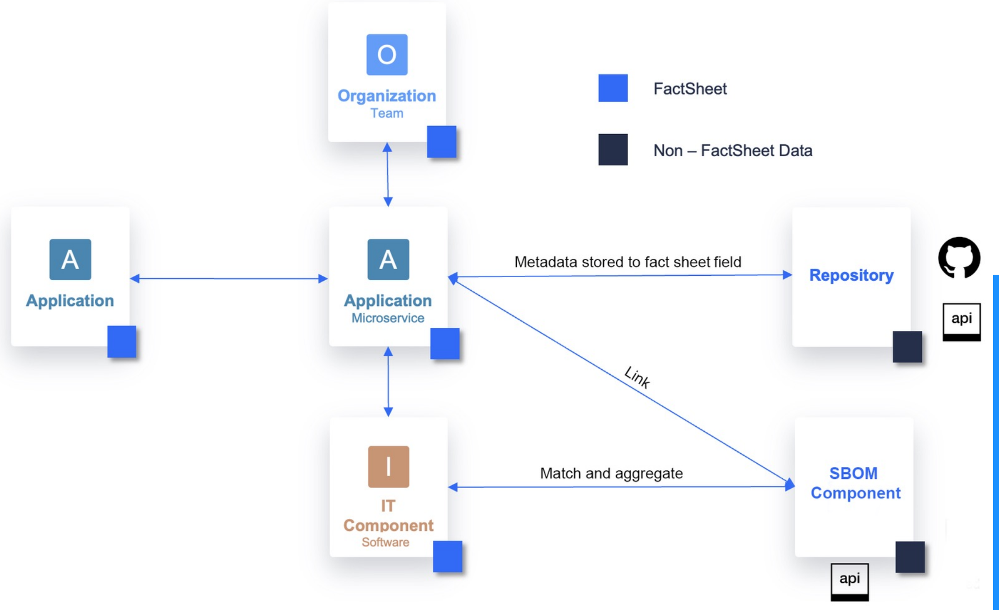
Self-Built Software Discovery is a feature offered by SAP LeanIX Technology Risk and Compliance that enables **automated discovery** of components of your in-house development such as **microservices and technology stacks**. This way you can connect the worlds of developers and leaders into **one source of truth.** The information from the repository will be used to link development information to applications and will improve data accuracy through **established relationships between microservices, business applications and teams.**

WHY?

Internally built software is becoming a **growing part of IT portfolios**, however, the **currently limited visibility** of these leads to **increased risks**: technological dispersion limits integration options, technical debt decreases IT resilience, and security vulnerabilities through unsupported technology frameworks increase. For all these reasons, it is important to include Self-Built Software in application portfolio and risk management assessments.

HOW?

This feature allows you to **automatically discover microservices** using the **manifest file** from your repository, and then link them to your respective business applications and Teams (Subtype Organization). This feature also integrates Software **Bill of Materials** (SBOMs) via the Self-Built Software Discovery API to associate them with discovered microservices. Then, the components of the SBOMs are compared to our reference catalog, and the IT components are automatically created and linked to the relevant microservices. For more info.



Recommendation #1

Use Cases – AI Governance and Adoption (Artificial Intelligence)

WHAT?

AI governance and adoption is about managing applications and technologies using artificial intelligence within an organization. AI governance involves establishing rules and guidelines to **mitigate risks and ensuring ethical and responsible use**. AI adoption, on the other hand, refers to embedding AI technologies into operations and workflows to improve efficiency and drive innovation. These practices enable organizations to use AI technologies responsibly and effectively, while maximizing the value of their investments.

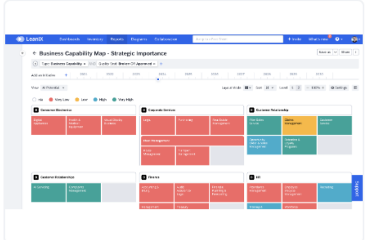
WHY?

AI governance is crucial for enterprise architecture as it **ensures AI technologies are aligned with business strategies**, which is critical for decision-making and strategic planning. Robust governance practices can provide a competitive advantage by maximizing the value of AI investments, ensuring ethical and regulatory compliance, and **effectively managing associated risks**.

HOW?

Start by **collecting data on the potential of AI, its current uses, and associated risks**. Use the capabilities of SAP LeanIX to evaluate this information and develop a risk mitigation strategy. Visualize and standardize AI technologies across your IT landscape and build strategic plans for transformations involving AI. By following these steps, you'll maximize the value of your AI investments while ensuring their compliance and positive impact on your organization. [See our step-by-step guidance](#).

Use Cases Extensions



AI Governance & Adoption Active

Optional Feature

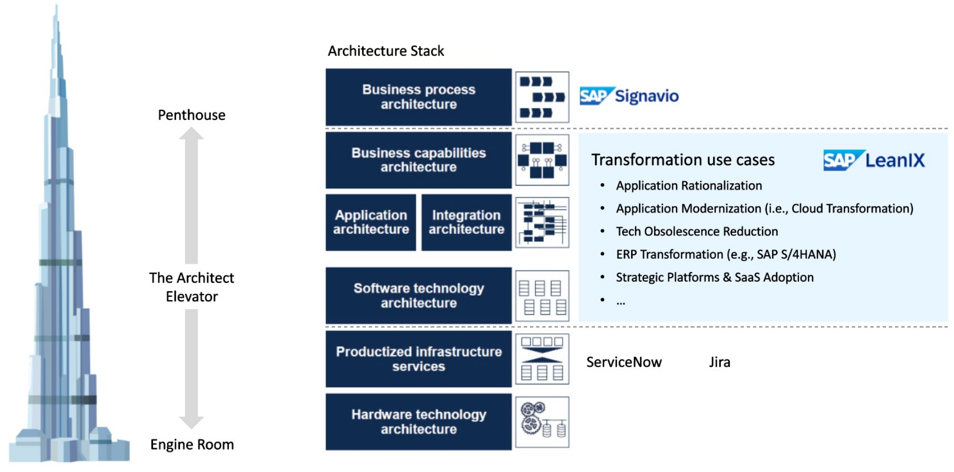
Recommendation #2
SAP LeanIX adoption - Enterprise Architecture at the Core of Strategic Decisions

Context

In a dynamic business environment, companies are looking to adapt quickly and innovate, putting **enterprise architecture at the center of strategic decisions**. Thus, adopting SAP LeanIX enables companies to increase transparency and governance of their enterprise architecture, by using executive dashboards or configuring dashboards per use case. These integrated and comprehensive views of systems **foster collaboration between IT and business functions.**

In addition, the focus on user adoption ensures that non-technical teams, likewise, are integrated into the process, maximizing the efficiency and strategic impact of the decisions put in place.

SAP LeanIX enables translation across layers



Source: The Software Architect Elevator, Gregor Hohpe

Best Practices

The successful adoption of SAP LeanIX as a strategic pillar of enterprise architecture decisions **is based on several key best practices**:

- Leverage **Dashboards**: Executives as well as use case based ones
- Ensure collaboration with business teams by **optimizing tool integration**
- Ensure **user engagement and adoption**
- Achieve **iterative progress** and continuous improvements
- Establishment of **monitoring** and evaluation tools

Recommendation #2

SAP LeanIX adoption – Executive Dashboard Architecture

WHAT?

The architecture executive dashboard in SAP LeanIX is a key feature that provides a strategic overview of the IT architecture, consolidating data across business systems and processes to support decision-making.

WHY?

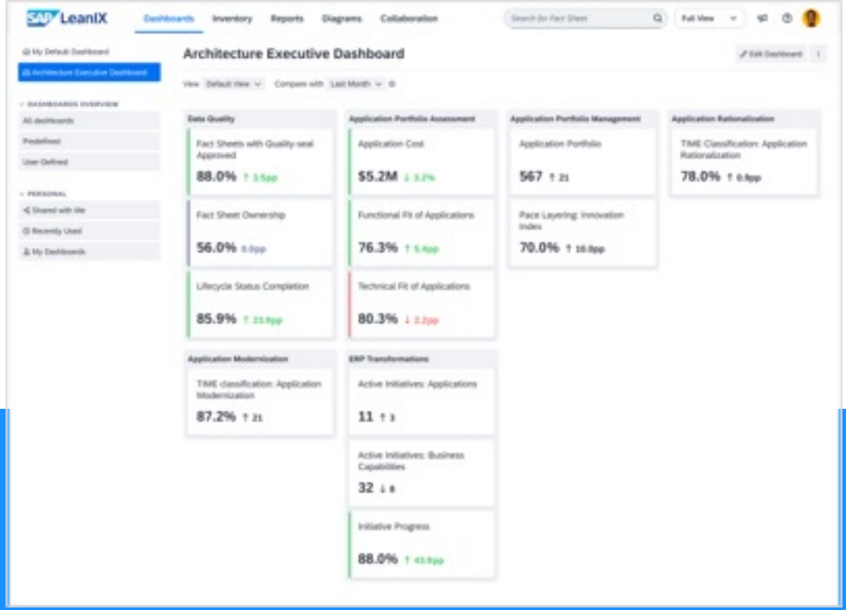
Adopting the executive dashboard is essential to optimize visibility and strategic decision-making within your IT architecture. This **enhances the ability to understand the state of systems and application portfolio**, provides **real-time KPIs and analytics** for better decision-making, and helps **align IT initiatives with business objectives**, supporting initiatives such as ERP modernization, data quality improvement, or cloud migration.

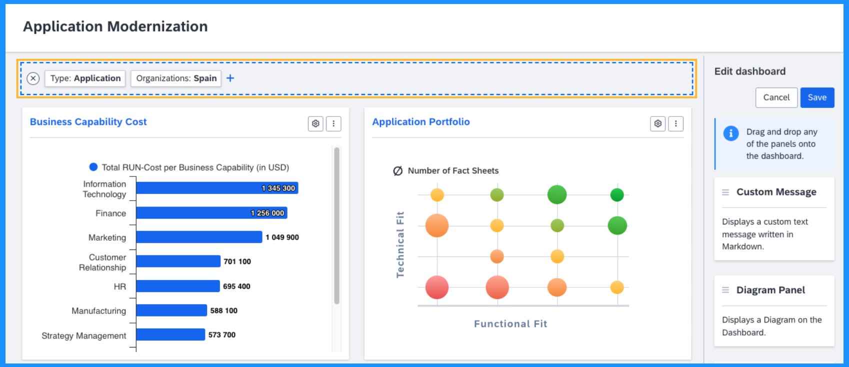
HOW?

To maximize the impact of the architecture executive dashboard, take a methodical approach that encompasses **personalization by defining views** (for instance: by organization, department, or market) and selecting KPIs that are [relevant] to the chosen segment. It is also recommended to create **dashboards per use cases**: for persona (CIO or Enterprise Architects), or for specific use cases (cost management, rationalization, business transformation, etc.).

It is also important to continuously **train** and educate users and administrators, and to ensure that **dashboards are integrated into decision-making processes** to ensure better collaboration between teams. Regular reviews ensure continuous improvements.

Some resources: The [Architecture Executive Dashboard] and its upcoming new features [(Roadmap)], as well as the [Use case specific dashboards].





Recommendation #2

SAP LeanIX adoption – Bringing Business Teams Together

WHAT?

Bringing business teams closer together means integrating enterprise architecture into strategic discussions, facilitating collaboration between IT and business.

WHY?

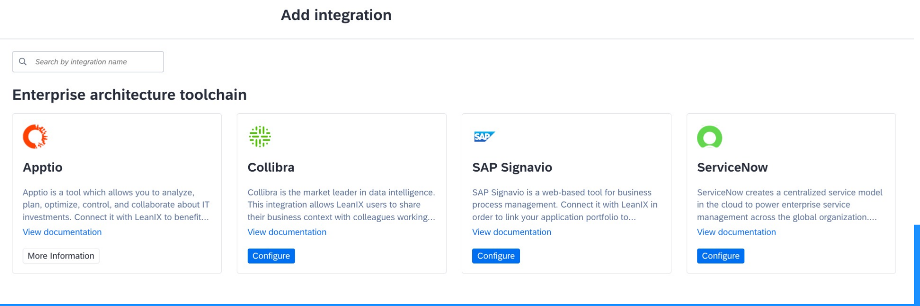
Driving close collaboration between IT and lines of business improves **strategic decision-making**, aligns goals, and drives innovation. By fostering **cross-functional collaboration**, you will open SAP LeanIX to other key business teams and stakeholders and drive consistent and innovative initiatives.

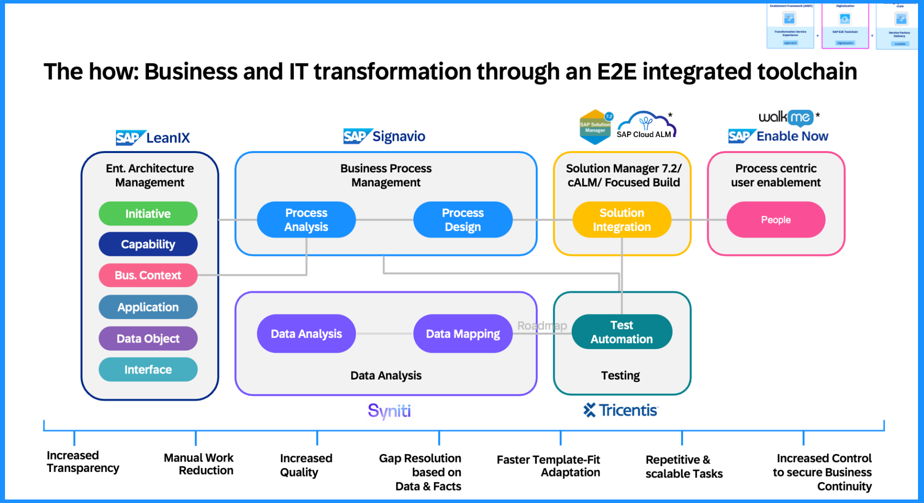
HOW?

Integration with SAP Signavio and import of processes (via imports or APIs) facilitate this synergy by making data accessible and actionable in a shared environment. The same applies to data management or technical components.

By adopting an integrated approach that fosters collaboration and integrations across solutions such as with Business Transformation Management (SAP LeanIX, SAP Signavio, and WalkMe), this ensures effective data management and supports consistent transformation initiatives across the organization.

The cultural dimension and **adaptability of teams strongly influence the adoption** of SAP LeanIX and their ability to collaborate effectively to maximize the impact of the solutions in place.
SAP's **Integrated Toolchain** aims to connect its solutions effectively to provide seamless management of business processes across the entire digital ecosystem (short version of our Integrated Toolchain and a more detailed presentation).





Recommendation #2

SAP LeanIX adoption – Users

WHAT?

Embedding SAP LeanIX at the core of organizational processes maximizes its use by users and strengthens its role in strategic discussions.

WHY?

Ensuring this commitment is important, to enable SAP LeanIX to become a catalyst that drives engagement and ensures successful product adoption across the organization.

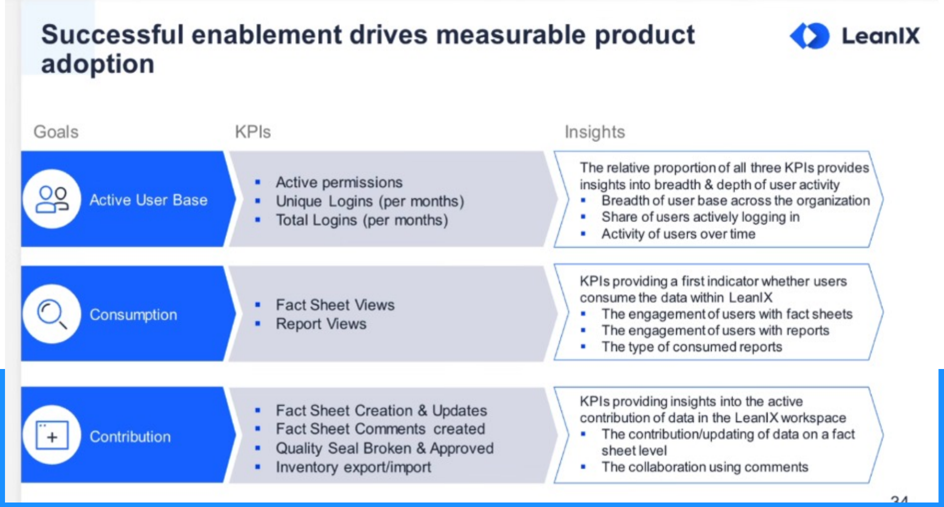
HOW?

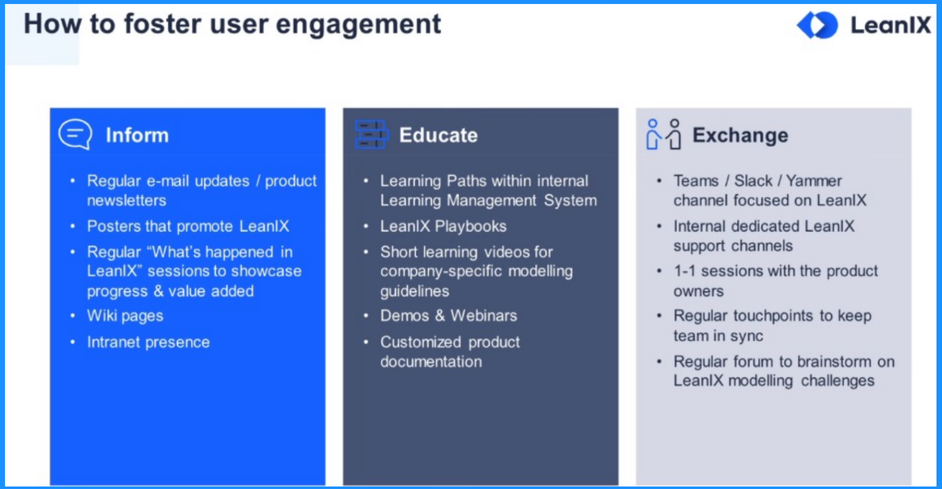
We can highlight several features that will make it easier to engage with users and share insights with management:

- **Measure adoption**: via KPIs to evaluate data quality, active user base, monitor consumption through consultations and contribution to factsheets, reports or diagrams, imports and exports … You can use your Shared 360 to monitor the usage and adoption of SAP LeanIX (*contact your CSM or customer success teams to access it)*

- **Keep users engaged** by focusing on the three pillars:
    - **Inform –** regularly distribute information to users (updates emails, newsletters, posters, office hours, etc …)
    - **Educate –** offer educational resources (tutorials, training sessions, playbooks, webinars, documentation, etc …)
    - **Engage –** encourage communication and experience sharing (Teams, Slack, dedicated channels, product sessions, alignment sessions, etc …)

Put **SAP LeanIX at the heart of your organization**, integrate it with your projects via Jira or Confluence, and development teams, among others.





Recommendation #3
Governance: calculations, integrations and architecture decisions

Context

As we have already pointed out several times, **IT governance** is an important topic to **ensure IT's efficiency, security, and alignment with business objectives**. Ungoverned or limited IT environments can result in significant risks such as fragmented tools, competing projects, technical debt, and security vulnerabilities.
These risks can in turn lead to increased **maintenance costs**, resource conflicts, **reduced agility**, and increased **vulnerability** to data breaches.

In this level 3, we will focus on calculations, integrations, and architecture, decisions and how these 3 capabilities can support the implementation of a successful governance.

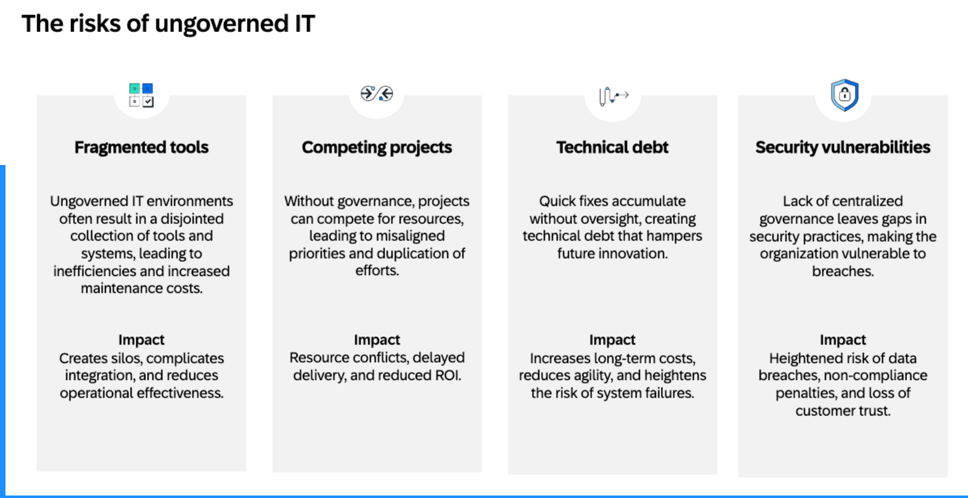
Best Practices

IT governance is enhanced by calculations, custom integrations, and architecture decisions, functionalities that all contribute to the efficiency and strategic alignment of IT systems.

- **Calculations** automate data updates, reduce errors, and provide an accurate view of the architecture for effective strategic decisions.

- **Custom integrations** connect systems to specific needs, consolidating data to improve operational consistency and reduce silo risks.

- **Architecture decisions** centralize architectural choices, fostering collaboration and visibility for a secure, high-performance IT framework.



Recommendation #2

Governance - Calculations

WHAT?

SAP LeanIX allows you to create **calculations to automatically populate** data in fact sheet fields using values from other fields in your inventory. You can define the calculation logic using JavaScript code, this provides the ability to automate the data population without the need for third-party solutions.

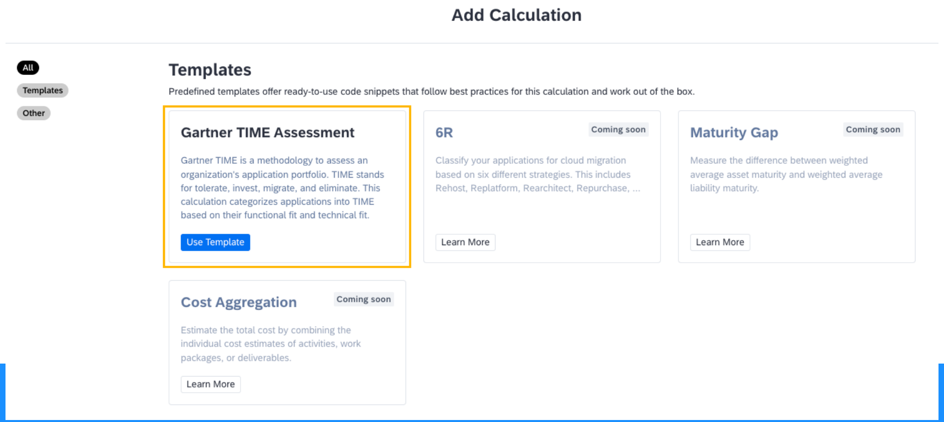
WHY?

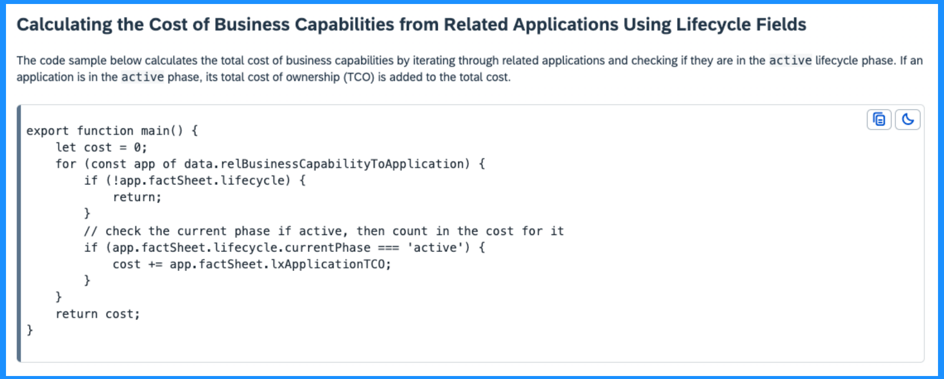
Automated calculations **improve data governance** by ensuring accurate, up-to-date information without manual intervention, reducing **human errors and increasing operational efficiency.** They also provide the flexibility to customize the calculation logic to the specific needs of the organization, while ensuring secure execution in the SAP LeanIX environment.

HOW?

Here are some recommendations for getting started with Calculations:

- **Configuration and writing**: From the administration, access predefined calculation templates or configure custom calculations (Javascript code).
- **Check and test**: Use the technical keys of the fields to ensure correct identification of the calculations, and test your calculations before activating them.
- **Enable and trace**: Enable calculations to be triggered during defined changes, and monitor the error log to detect and correct any issues.
- **Precautions and change management**: Avoid dependency loops (might block your calculations), and set automatically populated fields in "read-only" mode to prevent unwanted manual changes.





Calculating the Cost of Business Capabilities from Related Applications Using Lifecycle Fields

The code sample below calculates the total cost of business capabilities by iterating through related applications and checking if they are in the `active` lifecycle phase. If an application is in the `active` phase, its total cost of ownership (TCO) is added to the total cost.

```
export function main() {
    let cost = 0;
    for (const app of data.relBusinessCapabilityToApplication) {
        if (!app.factSheet.lifecycle) {
            return;
        }
        // check the current phase if active, then count in the cost for it
        if (app.factSheet.lifecycle.currentPhase === 'active') {
            cost += app.factSheet.lxApplicationTCO;
        }
    }
    return cost;
}
```

Recommendation #2
Governance - Customized Integrations

WHAT?

Governance with SAP LeanIX is facilitated by custom integrations, enabling efficient data import and export and workflow automation. Several integration technologies are available, including out-of-the-box integrations, GraphQL API, REST APIs, webhooks, and Integration API, which offer solutions tailored to the specific needs of each business.

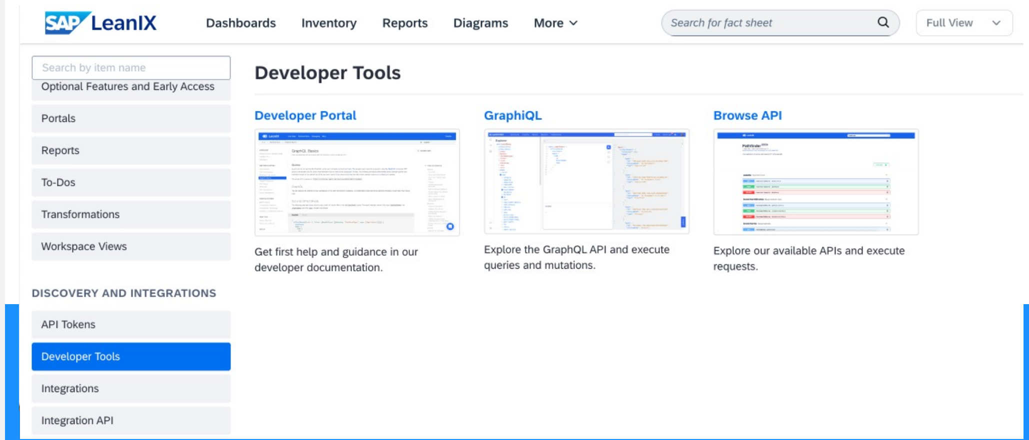
WHY?

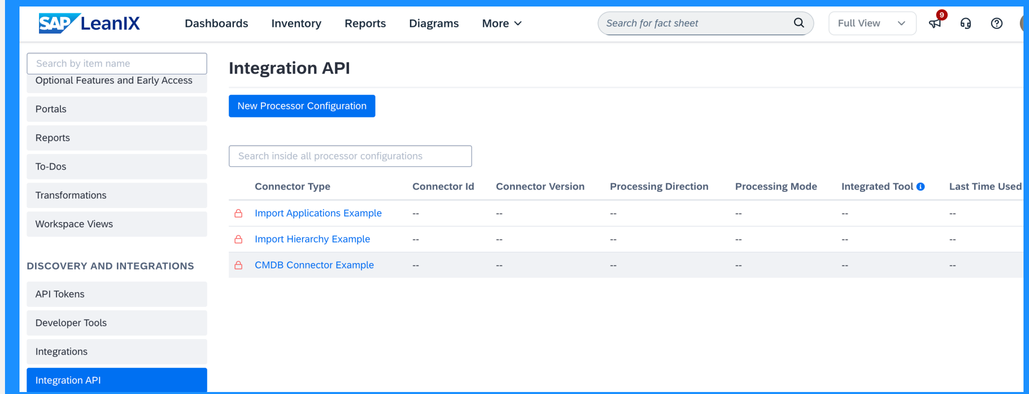
Customized integrations strengthen IT governance by optimizing the connection between external and internal systems, ensuring smooth and consistent data management. This helps minimize risks associated with ungoverned IT environments and aligns IT assets with strategic business objectives.

HOW?

Here are some clear and simple recommendations for a successful implementation of custom integrations with SAP LeanIX:

- **Define needs**: Identify the specific requirements for your integration.
- **Choose the right technology**: Select between GraphQL API, REST API, or out-of-the-box integrations as needed.
- **Set up and test**: Use SAP LeanIX tools to set up your integrations and check their effectiveness.
- **Optimize synchronization**: Adjust the timing of data synchronization to meet your requirements, using [webhook] events for real-time synchronizations where possible.
- **Use available resources**: Refer to SAP LeanIX [documentation] and seek guidance from Customer Success team as needed.





Recommendation #2

Governance - Architecture Decisions

WHAT?

The **Architecture Decisions** feature of SAP LeanIX is designed to **streamline and document architectural choices** within your IT landscape. This feature provides a centralized platform for **recording, managing, and communicating key architectural decisions**, ensuring consistency and transparency across your organization.

WHY?

Effective governance relies on making and documenting architectural decisions, but the decision-making process can often seem fragmented and chaotic.
Within the framework of architectural decisions, business architects face **several obstacles**, such as **scattered decision recordings, opaque decision making, and risky practices** (see image on the right).

HOW?

Here are some good governance practices:
Create an Architecture Council (ARB) with regular meetings, establish clear procedures for submitting proposed decisions, enable asynchronous decision-making, and ensure that all processes adhere to legal and compliance standards.

The SAP LeanIX **Architecture Decisions** feature will enable you **to get a centralized view, standardized documentation, and improved visibility and collaboration** through the features that can be found [here](#) in our documentation.

Enterprise architect's hurdles in making effective architecture decisions